

Individual Capstone Assessment Essay

Our senior design project, Project “PrintNet”, is a web-based 3D Printing Lab Management Platform designed to modernize the UC 3D Print Farm. The platform draws inspiration from 3D Print OS but is customized to fit UC’s environment, including features such as an in-browser slicer, self-help tutorials, printer maintenance support, and UC email authentication with role-based access. Students will benefit from a streamlined submission process, while lab managers will be able to approve jobs, adjust slicer settings, and conduct maintenance tasks on printers ranging from Maker Gear Series 2 to Bambu Lab models. Administrators will have oversight capabilities, including toggling cost visibility and managing lab access. From my perspective as a Computer Science major, this project represents the integration of my academic studies, my professional experiences, and my leadership in UC’s 3D Design & Automation Club.

My coursework at UC has prepared me well for this challenge by building a foundation across algorithms, systems, software engineering, and networking. In CS 1021C: Computer Science I and CS 2028C: Data Structures, I developed skills in programming fundamentals and efficient data handling—knowledge that will help in structuring print queues and workflows. CS 2011: Introduction to Computer Systems provided me with a deeper understanding of low-level computation and operating system

concepts, which are important for handling concurrent requests and hardware integration.

CS 2071: Discrete Structures strengthened my ability to reason about logic and formal systems, essential when building reliable authentication and access control mechanisms.

Advanced courses such as CS 4071: Design and Analysis of Algorithms and EECE 4029: Operating Systems & Systems Programming will be directly applied when optimizing scheduling across multiple printers. Likewise, CS 4033: Database Design and Development and CS 4065: Computer Networks gave me the tools to design robust schemas, build relational workflows, and ensure reliable communication between the frontend, backend, and printer hardware. I also gained valuable experience in EECE 3093C: Software Engineering, where I worked in teams to design, test, and deliver software projects—preparing me to collaborate effectively with my teammates on this capstone.

Equally important are the experiences I gained through my co-op rotations at GE Appliances (SmartHQ / Connected Home Cloud). There, I worked as a Software and Cloud Engineer on AWS microservices, developing expertise with DynamoDB, Lambda, Kinesis, and CloudFormation. I regularly troubleshoot Maven builds, dependency conflicts, and CI/CD pipelines—skills that will directly apply to setting up backend infrastructure and deployment pipelines for PrintNet. I also worked on embedded software and mobile integration, which taught me how to bridge user-facing interfaces with physical devices—similar to linking slicer outputs and printer controllers in this project. Beyond technical work, my co-op emphasized soft skills such as communication, documentation, and agile collaboration. These experiences gave me the ability to not only write code but also ensure

that the project is well-managed and can be transitioned to future lab managers after my graduation.

I am especially motivated to work on this project because I have direct experience managing UC's 3D printing lab and have seen firsthand how challenging it is to keep workflows efficient at scale. Existing tools like OctoPrint are useful for maintenance but lack the tracking and eligibility features needed in a student environment. Project PrintNet provides an opportunity to design a system that is not only technically strong but also tailored for the daily realities of UC students and managers. My approach will be modular: a student-facing portal with integrated slicing and help tools, a backend for job tracking and authentication, and a flexible workflow system powered by Airtable or an internal solution. I will also prioritize compatibility with both legacy and modern printers to future-proof the system.

The expected outcome is a platform that is reliable, scalable, and user-friendly. Students should be able to submit jobs quickly and track their status, managers should have the tools to monitor and maintain printers effectively, and administrators should be able to oversee access, costs, and analytics. My personal contributions will focus on the critical integrations: authentication, printer compatibility, workflow management, and backend deployment. I will evaluate my success by testing system performance under multiple users, verifying maintenance features on different printer models, and ensuring

that role-based access functions securely. Ultimately, I will know I have succeeded when the system runs smoothly without constant oversight, enabling the UC 3D Print Farm to scale efficiently for future students and managers.